21-242 Recitation 12

Dimension — Q5. Dimension of a subspace

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

If $V \subseteq W$ is a subspace, $\dim(V) \leq \dim(W)$.

Course-level solution

If $\dim W = \infty$ there is nothing to prove. If $\dim W = n < \infty$, a basis of V is a linearly independent list in W, hence has length at most n by Q1.
(If $\dim V = \dim W < \infty$ then any basis of V is an independent list of length $\dim W$ in W, hence a basis of W, so $V = W$.)